

Question ID: 29ed5d39

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$$p = 20 + \frac{16}{n}$$

The given equation relates the numbers p and n , where n is not equal to 0 and $p > 20$. Which equation correctly expresses n in terms of p ?

Answer

- A. $n = \frac{p-20}{16}$
- B. $n = \frac{p}{16} + 20$
- C. $n = \frac{p}{16} - 20$
- D. $n = \frac{16}{p-20}$

Correct Answer: D

Rationale

Choice D is correct. To express n in terms of p , the given equation must be solved for n . Subtracting 9 from both sides of the given equation yields $p - 9 = \frac{14}{n}$. Since n is not equal to 0, multiplying both sides of this equation by n yields $(p - 9)(n) = 14$. It's given that $p > 9$, which means $p - 9$ is not equal to 0. Therefore, dividing both sides of $(p - 9)(n) = 14$ by $(p - 9)$ yields $\frac{(p-9)(n)}{p-9} = \frac{14}{p-9}$, or $n = \frac{14}{p-9}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.